

ETL PIPELINE PROGRESS REPORT

Sales Dataset – Transformation Decisions & Data Warehousing Design

GitHub Documentation / Project Report

1. Project Overview

This report documents the ETL work completed for the Sales Dataset in Google Colab. It focuses on the decisions made during transformation and the data-warehousing structure selected for the analytical model.

2. Source Dataset

The project uses three related CSV datasets:

- sales.csv – transaction-level sales data
- customers.csv – customer information
- products.csv – product information and product price data

3. Transformation Decisions

The transformation stage was designed to preserve valid transaction records while standardizing the data and preparing it for later integration.

3.1 Column Name Standardization

Source column names were converted to a consistent snake_case naming convention. For example: OrderID → order_id, OrderDate → order_date, CustomerID → customer_id, ProductID → product_id, Quantity → quantity and UnitPrice → unit_price.

3.2 Date Transformation

OrderDate was converted to a pandas datetime type. Invalid date values are coerced to NaT so that they can be detected and handled as data-quality issues.

3.3 Numeric Transformation

Quantity was converted to a numeric data type. UnitPrice was also prepared as a numeric field, but the source sales dataset contained missing UnitPrice values.

3.4 Missing UnitPrice Decision

The most important transformation decision was not to reject sales records only because UnitPrice was missing. UnitPrice was completely missing in the raw sales data. Since the products dataset contains the corresponding product price (price_usd), the missing price was intentionally left as NaN during transformation and scheduled for enrichment during the integration stage.

Field	Current State	Decision
unit_price	1,000 missing	Keep temporarily; enrich from products
sales_amount	1,000 missing	Calculate after price enrichment
total_price	1,000 missing	Calculate after price enrichment

3.5 Financial Calculation Decision

sales_amount was defined as quantity × unit_price. Because unit_price was missing at the transformation stage, sales_amount and total_price were also temporarily missing. This avoids creating incorrect financial values from an unreliable price.

3.6 Redundant Column Handling

A redundant salesamount column was removed before calculating the standardized sales_amount field. This prevents duplicate/conflicting representations of the same measure.

3.7 Record Rejection Rule

Rows were rejected only when critical transaction information was missing: customer_id, product_id, quantity or order_date. Missing UnitPrice alone was not treated as a reason for rejection.

4. Data Quality Results

- Total transformed sales rows: 1,000
- Rejected rows: 0
- Unique order IDs: 1,000
- Unique customer IDs: 638
- Unique product IDs: 614
- Duplicate order IDs: none identified
- Data-quality status at this stage: WARNING, because financial fields are intentionally pending enrichment

5. Data Warehousing Approach

For the data warehouse, a Star Schema approach was selected. The central fact table stores measurable sales transactions, while dimension tables provide descriptive information used for analysis.

5.1 Fact Table – fact_sales

The planned central fact table is fact_sales. It contains transaction measures and foreign keys.

- order_id – transaction identifier
- customer_key – foreign key to dim_customer
- product_key – foreign key to dim_product
- date_key / date_id – foreign key to dim_date
- quantity – sales quantity
- unit_price – product unit price after enrichment
- sales_amount – calculated sales measure
- total_price – transaction total

5.2 Dimension Table – dim_customer

dim_customer is used to store customer-related descriptive attributes. A surrogate customer_key is used for warehouse relationships, while customer_id remains the business/source identifier.

5.3 Dimension Table – dim_product

dim_product is used to store product-related descriptive attributes. A surrogate product_key is used for warehouse relationships, while product_id remains the business/source identifier. Product price is also the intended source for resolving the missing UnitPrice values in sales.

5.4 Dimension Table – dim_date

dim_date was successfully created from the unique order dates. It contains calendar attributes to support time-based analysis.

- 609 rows
- 12 columns
- Primary key: date_id
- Date range: 2023-01-01 to 2025-06-28
- Attributes: full_date, day_of_week, day_of_month, day_of_year, week_of_year, month, month_num, quarter, year, is_weekend, is_holiday and season

6. Why Star Schema Was Selected

- Separates measurable sales transactions from descriptive business information.
- Makes sales analysis by customer, product and date easier.
- Supports aggregation such as total sales, quantity and order-level analysis.
- Reduces repeated descriptive data in the central transaction table.
- Provides a clear structure for reporting and visualization.

7. Current Warehousing Status

At the current documented stage, dim_date has been created and validated successfully. The customer and product dimensions and fact_sales are the next warehouse construction steps. Product-price enrichment must occur before final financial measures are populated.

Warehouse Component	Purpose	Status
dim_date	Calendar/time analysis	Completed – SUCCESS
dim_customer	Customer attributes	Next stage
dim_product	Product attributes + price	Next stage
fact_sales	Central sales measures + foreign keys	Next stage

8. Key ETL Decision Summary

- Preserve valid sales records instead of dropping all rows because UnitPrice is missing.
- Enrich UnitPrice from the product dataset during integration.
- Calculate financial measures only after reliable product prices are available.
- Use standardized snake_case column names.
- Use datetime for order_date.
- Use numeric types for quantity and price/measure fields.
- Use a Star Schema for the warehouse.
- Use surrogate keys for warehouse dimensions and foreign-key relationships in fact_sales.
- Use a dedicated date dimension for time-based reporting.

9. Remaining ETL / Warehouse Work

- Product price enrichment
- Customer integration
- Product integration
- Creation and validation of dim_customer
- Creation and validation of dim_product
- Creation of fact_sales
- Final analytics and KPIs
- Business analysis and visualizations
- Loading processed warehouse tables
- Final validation and ETL summary

10. Tools Used

- Python
- Pandas
- NumPy
- Matplotlib
- Google Colab
- Jupyter Notebook

11. Conclusion

The transformation stage has established a clean and consistent foundation for the warehouse. The key design decision was to retain all valid sales transactions even though UnitPrice is missing in the source sales table, because the product dataset can provide the required price during integration. A Star Schema consisting of fact_sales, dim_customer, dim_product and dim_date has been selected for the data warehouse. The date dimension is already successfully created and validated; the remaining warehouse tables will be completed in the subsequent integration and loading stages.